

Factoring Algebraic Expressions (Part III)

Factoring Trinomials of the Form ax^4+bx^2+c .

- ① $x^4+13x^2+40 = (x^2+5)(x^2+8)$
- ② $y^4-4y^2-45 = (y^2-9)(y^2+5) = (y-3)(y+3)(y^2+5)$
- ③ $z^4+4z^2-45 = (z^2+9)(z^2-5)$
- ④ $a^4-13a^2+36 = (a^2-9)(a^2-4) = (a-3)(a+3)(a-2)(a+2)$
- ⑤ $x^4+5x^2-14 = (x^2+7)(x^2-2)$
- ⑥ $y^4-8y^2+16 = (y^2-4)(y^2-4) = (y+2)(y-2)(y+2)(y-2)$
- ⑦ $z^4-5z^2+4 = (z^2-1)(z^2-4) = (z+1)(z-1)(z-2)(z+2)$
- ⑧ $x^4-2x^2-15 = (x^2-5)(x^2+3)$
- ⑨ $y^4+4y^2-5 = (y^2+5)(y^2-1) = (y^2+5)(y-1)(y+1)$
- ⑩ $z^4+7z^2-18 = (z^2+9)(z^2-2)$
- ⑪ $x^4+8x^2+12 = (x^2+2)(x^2+6)$
- ⑫ $y^4-5y^2+6 = (y^2-2)(y^2-3)$
- ⑬ $z^4+8z^2-9 = (z^2+9)(z^2-1) = (z^2+9)(z+1)(z-1)$
- ⑭ $x^4-18x^2+81 = (x^2-9)(x^2-9) = (x-3)(x+3)(x-3)(x+3) = (x-3)^2(x+3)^2$
- ⑮ $y^4-2y^2+1 = (y^2-1)(y^2-1) = (y-1)(y+1)(y-1)(y+1) = (y-1)^2(y+1)^2$
- ⑯ $z^4+2z^2+1 = (z^2+1)(z^2+1)$

$$⑰ 2x^4+11x^2+5 = (2x^2+1)(x^2+5)$$

$$⑱ 3y^4+10y^2-25 = (3y^2-5)(y^2+5)$$

$$⑲ 3z^4-5z^2-12 = (3z^2+4)(z^2-3)$$

$$⑳ 8x^4+2x^2-3 = (2x^2-1)(4x^2+3)$$

$$㉑ 9y^4-25y^2+16 = (9y^2-16)(y^2-1) = (3y+4)(3y-4)(y-1)(y+1)$$

Factoring by Grouping

$$㉒ x^3+4x^2+4x+16 = x^2(x+4)+4(x+4) = (x+4)(x^2+4)$$

$$㉓ 6y^3-10y^2+9y-15 = 2y^2(3y-5)+3(3y-5) = (3y-5)(2y^2+3)$$

$$㉔ z^5-z^4-81z+81 = z^4(z-1)-81(z-1) = (z-1)(z^4-81) = (z-1)(z^2-9)(z^2+9) = (z-1)(z-3)(z+3)(z^2+9)$$

$$㉕ x^3-5x^2-9x+45 = x^2(x-5)-9(x-5) = (x-5)(x^2-9) = (x-5)(x+3)(x-3)$$

$$㉖ 12y^4-6y^3+2y-1 = 6y^3(2y-1)+2y-1 = (2y-1)(6y^3+1)$$

$$㉗ 20x^3+12x^2+5x+3 = 4x^2(5x+3)+1(5x+3) = (5x+3)(4x^2+1)$$

$$\begin{aligned} \textcircled{28} \quad x^3 - 3x^2 - 4x + 12 &= x^2(x-3) - 4(x-3) \\ &= (x-3)(x^2-4) = \boxed{(x-3)(x-2)(x+2)} \end{aligned}$$

$$\begin{aligned} \textcircled{29} \quad 5z^3 - 35z^2 - z + 7 &= 5z^2(z-7) - 1(z-7) \\ &= \boxed{(z-7)(5z^2-1)} \end{aligned}$$

$$\begin{aligned} \textcircled{30} \quad x^3 - 2x^2 - x + 2 &= x^2(x-2) - 1(x-2) \\ &= (x-2)(x^2-1) = \boxed{(x-2)(x-1)(x+1)} \end{aligned}$$

$$\begin{aligned} \textcircled{31} \quad 2z^6 - 18z^5 - z + 9 &= 2z^5(z-9) - 1(z-9) \\ &= \boxed{(z-9)(2z^5-1)} \end{aligned}$$

$$\begin{aligned} \textcircled{32} \quad x^3 - 9x^2 - x + 9 &= x^2(x-9) - 1(x-9) \\ &= (x-9)(x^2-1) = \boxed{(x-9)(x+1)(x-1)} \end{aligned}$$

$$\begin{aligned} \textcircled{33} \quad 4y^3 + 12y^2 - y - 3 &= 4y^2(y+3) - 1(y+3) \\ &= (y+3)(4y^2-1) = \boxed{(y+3)(2y-1)(2y+1)} \end{aligned}$$

$$\begin{aligned} \textcircled{34} \quad z^5 + 4z^4 - z - 4 &= z^4(z+4) - 1(z+4) \\ &= (z+4)(z^4-1) = \boxed{(z+4)(z^2-1)(z^2+1)} \\ &= \boxed{(z+4)(z-1)(z+1)(z^2+1)} \end{aligned}$$

$$\begin{aligned} \textcircled{35} \quad 5a^3 - 4a^2 + 35a - 28 &= \\ &= a^2(5a-4) + 7(5a-4) \\ &= \boxed{(5a-4)(a^2+7)} \end{aligned}$$

$$\begin{aligned} \textcircled{36} \quad x^3 + 10x^2 - 4x - 40 &= x^2(x+10) - 4(x+10) \\ &= (x+10)(x^2-4) = \boxed{(x+10)(x-2)(x+2)} \end{aligned}$$

$$\begin{aligned} \textcircled{37} \quad x^3 - 15x^2 - 3x + 45 &= x^2(x-15) - 3(x-15) \\ &= \boxed{(x-15)(x^2-3)} \end{aligned}$$

$$\begin{aligned} \textcircled{38} \quad 4x^3 + 16x^2 - 9x - 36 &= 4x^2(x+4) - 9(x+4) \\ &= (x+4)(4x^2-9) \\ &= \boxed{(x+4)(2x-3)(2x+3)} \end{aligned}$$